Vet Kita – Quick Guide & Technical Overview

IT 203 · Group B · BSIT 2H-G2

1. How to Run It

- **Database Setup:** Run schema.sql in MySQL Workbench to create the vetclinic_db_2 database and tables, then execute seed.sql.
- **Credentials Configuration:** Verify the database connection strings (DB_URL, DB_USER, DB_PASSWORD) in MySqlStorage.java.
- **Dependencies:** Ensure the MySQL Connector/J JAR is placed in the lib/ folder and linked in your IDE (IntelliJ / NetBeans) classpath.
- **Execution:** Run vetclinic.Main.java and log in with default admin credentials (admin / admin123).

2. Code Organization & Package Breakdown

Package	Key Classes	Primary Responsibility
model	Animal, Dog, Cat, Livestock, Staff, Veterinarian, VetAssistant, Owner, TreatmentRecord, Interfaces (Describable, Treatable)	Domain entities, polymorphism, and business validation rules.
service	ClinicManager, AuthService	Application state management, search filtering, ID generation, and SHA-256 authentication.
storage	MySqlStorage, Storable	Relational data persistence engine implementing JDBC transactions and automated schema migration.

exception	AuthenticationException, StorageException	Custom checked exceptions for runtime error propagation.
ui	MainFrame, DashboardPanel, AnimalPanel, StaffPanel, TreatmentPanel, modal forms (AnimalFormDialog, etc.)	Swing window layouts, custom UI components, and interactive dialogs.

Detailed Package Breakdown:

- **model**: Abstract Animal base class extended by Dog, Cat, and Livestock (polymorphically overriding care notes and intervals). Abstract Staff base class extended by Veterinarian and VetAssistant (enforcing treatment execution permissions). Owner, TreatmentRecord, and the Describable / Treatable interfaces.
- **service**: ClinicManager holding in-memory lists, keyword search filters, ID generation logic, and checkup/follow-up notification triggers; AuthService handling cryptographic SHA-256 password hashing.
- **storage**: MySqlStorage implementing Storable, utilizing PreparedStatement guards, atomic transactions, and automated schema migration checks (ensureCreatedAtColumn).
- **ui**: MainFrame housing a custom sidebar and header, individual tab panels (DashboardPanel, AnimalPanel, StaffPanel, TreatmentPanel), and undecorated modal dialog forms.

3. Architecture & Execution Flow

- **Startup Lifecycle:** SwingUtilities.invokeLater() launches Main, which instantiates MySqlStorage, injects it into ClinicManager, loads all records into memory via loadAll(), seeds default staff if empty, and opens LoginFrame.
- **In-Memory Processing:** Viewing data, keyword searching across multiple attributes, species/role filtering, and computing dashboard metrics happen instantly in memory via ClinicManager to eliminate redundant database round-trips.
- **State Changes & Transactions:** Modifying records follows a strict pipeline: validate input via model setters, update in-memory lists, execute atomic JDBC transactions in MySqlStorage (saveAll()), and handle errors gracefully via JOptionPane.

4. Advanced Implementations Beyond Core Lessons

What	Where	Why We Used It
Custom Undecorated Windows & Dragging Listeners	ui (*Dialog.java)	To build a modern, sleek desktop UI layout with custom headers and window-dragging mechanics.
Advanced Swing Layout Managers	ui (MainFrame, DashboardPanel, *Panel)	Leveraged BorderLayout for panel scaffolding, GridLayout for equal-width summary cards, GridBagLayout for responsive form scaling, BoxLayout for vertical stack flows, and CardLayout for dynamic view switching (empty states vs. active data panels).

Dependency Inversion via Interfaces	storage (Storable.java), service (ClinicManager.java)	Decoupling the persistence layer by plugging MySqlStorage into ClinicManager through the Storable contract.
Polymorphic Model Execution	model (Staff.treat(), Animal.getCareNote())	Delegating business logic dynamically based on staff role privileges and animal subtypes.
SHA-256 Cryptographic Hashing	service (AuthService.java)	Securing administrator credential verification against plaintext exposure.
Automated Schema Migration	storage (MySqlStorage.ensureCreatedAtColumn())	Dynamically patching missing database columns (created_at) on initial runtime startup.
High-DPI Crisp Icon Scaling	util (ImageUtil.java, CrispIcon)	Rendering sharp UI graphics and icons cleanly across scaled Windows high-DPI displays.